

Advanced Persistent Threat (APT) Simulation using Splunk & Sysmon

INDEX

1.	Abstract
2.	Introduction
3.	Objective
4.	System Architecture
5.	Implementation
6.	Attack Simulation
7.	Detection Techniques
8.	Dashboards
9.	Results
10.	Challenges
11.	Conclusions
12.	Reference

Title Page

Project Title: Advanced Persistent Threat (APT) Simulation using Splunk & Sysmon

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Course:- Cybersecurity

Abstract

This project focuses on simulating Advanced Persistent Threat (APT) attacks and detecting them using a Security Information and Event Management (SIEM) system. A lab environment was created using Splunk, Sysmon, and Windows to monitor system activity and analyze logs. Various attack techniques such as PowerShell execution, encoded commands, persistence through registry keys, and network communication were simulated. These activities were captured and analyzed using Splunk, where detection rules and alerts were created. The project demonstrates how real-world cyber attacks can be detected using log analysis and monitoring tools, providing practical knowledge of cybersecurity operations.

2. Introduction

Cybersecurity is the practice of protecting systems, networks, and data from digital attacks. With the increasing number of cyber threats, organizations require advanced monitoring systems to detect malicious activities.

An Advanced Persistent Threat (APT) is a type of cyber attack where an attacker gains unauthorized access to a system and remains undetected for a long period. These attacks are complex and involve multiple stages such as execution, persistence, and communication.

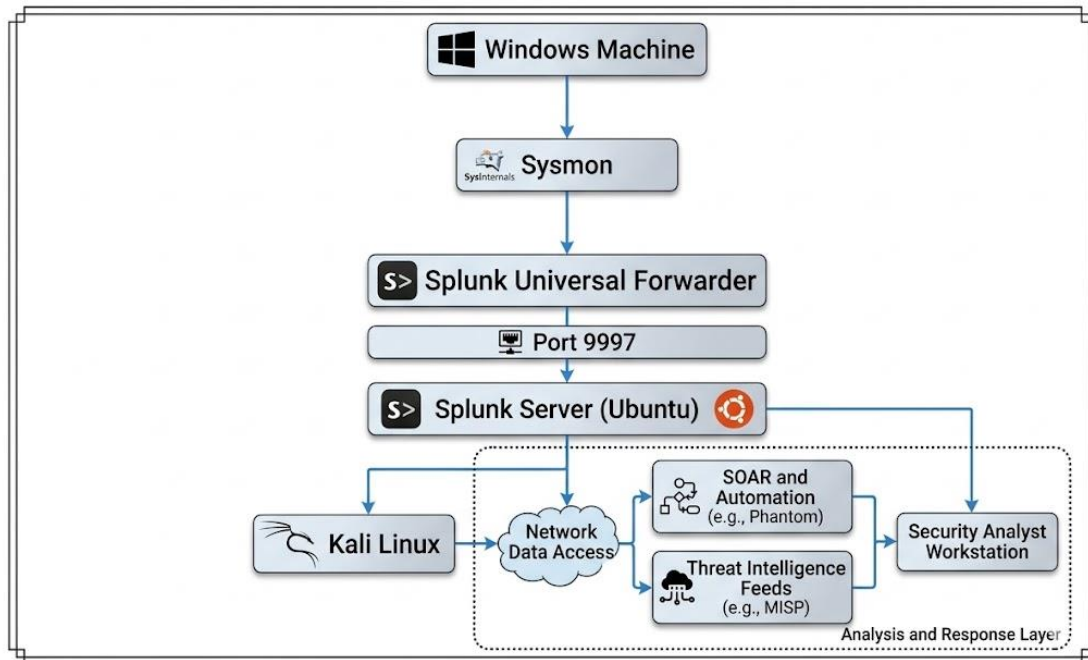
To detect such threats, organizations use SIEM tools like Splunk, which collect and analyze logs from different systems. This project aims to simulate APT behavior and detect it using Splunk and Sysmon, providing hands-on experience in real-world cybersecurity scenarios.

3.Objectives

- To build a SIEM-based monitoring system using Splunk
- To simulate attacker behavior in a controlled lab environment
- To detect suspicious activities using system logs
- To understand real-world cyber attack techniques
- To design a simulation framework that emulates the behavior of APT actors using open-source offensive security tools.
- To create a secure environment for performing and analyzing multi-stage attacks.
- To generate detection rules and threat intelligence indicators for defensive teams.
- To demonstrate end-to-end attack detection, logging, and mitigation within a lab setup.

4. System Architecture

The architecture of the system is as follows:



- Windows machine generates logs
- Sysmon captures detailed system activity
- Universal Forwarder sends logs to Splunk
- Splunk Server analyzes and visualizes logs

5.Tools and Technologies Used

◆ **Splunk Enterprise**

Used as a SIEM tool to collect, analyze, and visualize logs.

◆ **Sysmon**

A Windows system monitoring tool that records detailed system activity such as process creation, network connections, and registry changes.

◆ **Splunk Universal Forwarder**

Used to send logs from Windows machine to Splunk server.

◆ **Windows OS**

Used as the target system where activities and attacks are simulated.

◆ **Ubuntu Server**

Used to host the Splunk server.

◆ **Kali Linux**

Used to simulate attack scenarios such as network scanning

6.Implementation

Step 1: Setup Splunk Server

Splunk was installed on Ubuntu server and configured to receive data on port 9997.

```
user@admin:/opt/splunk/bin$ ./splunk status
splunkd is running (PID: 4133).
splunk helpers are running (PIDs: 4134 4440 4444 4551 4695 4957 5795 5807 5815 5817 42961 42967).
user@admin:/opt/splunk/bin$ _
```

Figure 1: Splunk Services Running on Ubuntu Server

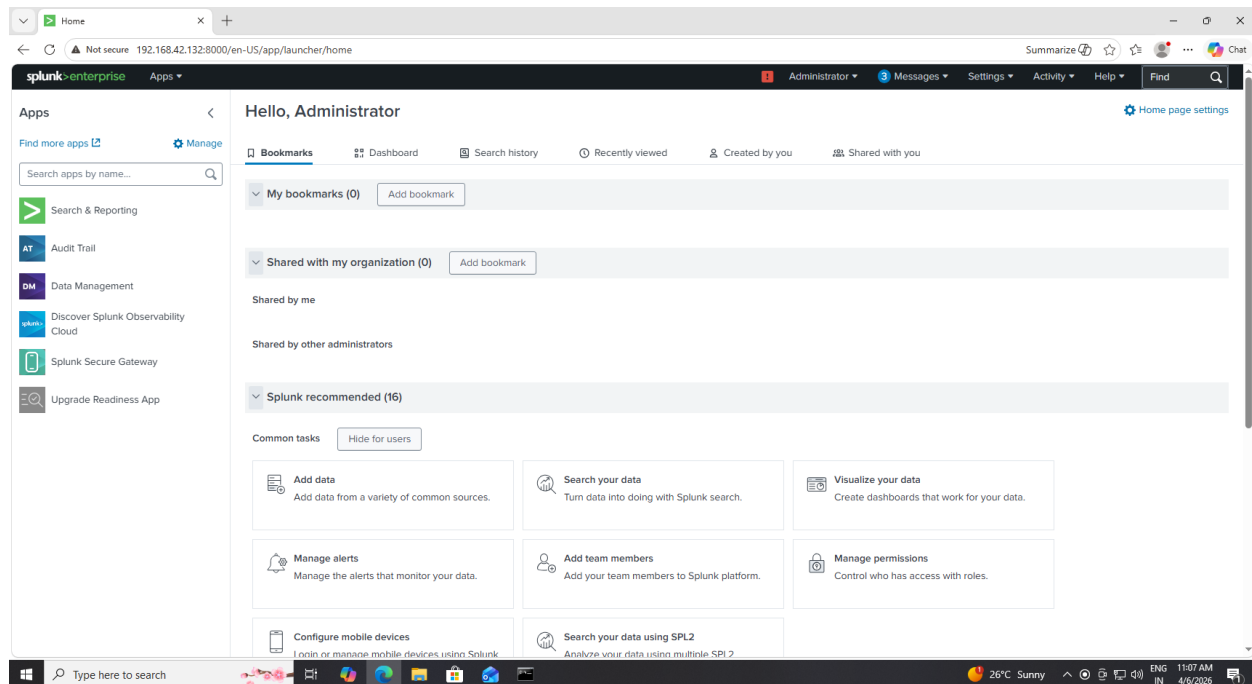


Figure 2: Splunk Web Interface Successfully Accessed

The above figure shows that the Splunk services are running successfully on the Ubuntu server.

The Web Interface Confirms That the Splunk server is accessible and ready to receive logs.

Step 2: Install Universal Forwarder

Splunk Universal Forwarder was installed on the Windows machine to send logs.

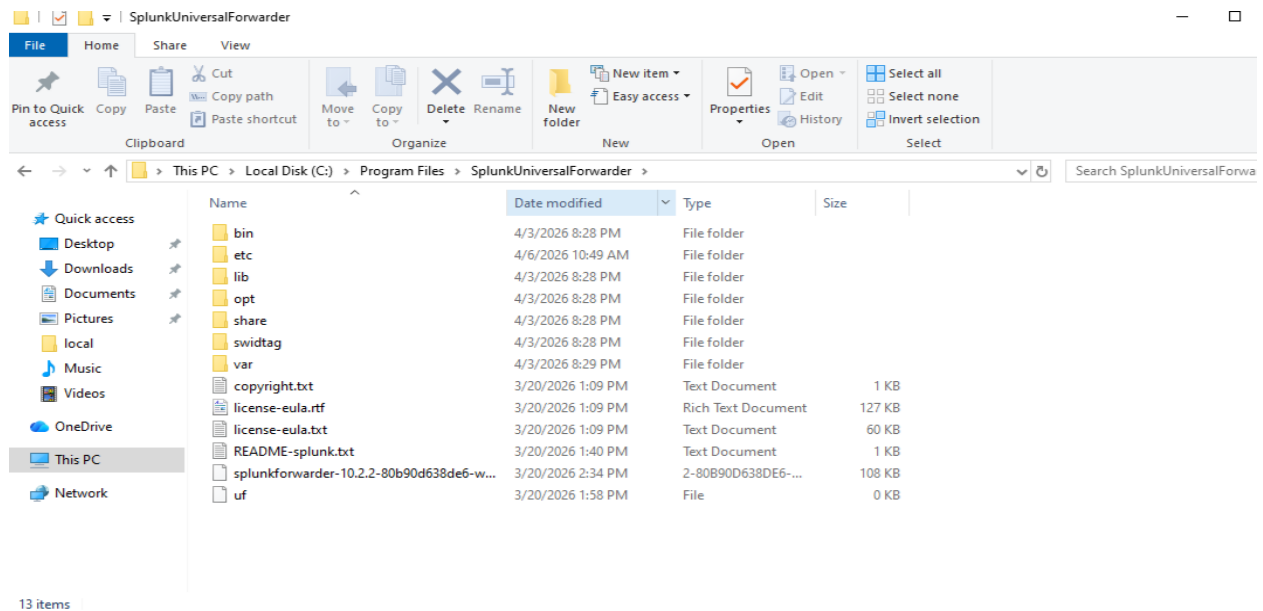


Figure 3: Splunk Universal Forwarder Installed Successfully on Windows

```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>sc query splunkforwarder

SERVICE_NAME: splunkforwarder
        TYPE               : 10  WIN32_OWN_PROCESS
        STATE                : 4   RUNNING
                        (STOPPABLE, NOT_PAUSABLE, ACCEPTS_SHUTDOWN)
        WIN32_EXIT_CODE       : 0    (0x0)
        SERVICE_EXIT_CODE   : 0    (0x0)
        CHECKPOINT           : 0x0
        WAIT_HINT            : 0x0

C:\Windows\system32>
```

Figure 4: Splunk Universal Forwarder Service Running

```
Command Prompt
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\admin>cd "C:\Program Files\SplunkUniversalForwarder\bin"

C:\Program Files\SplunkUniversalForwarder\bin>splunk start

Splunk> Another one.

Checking prerequisites...
  Checking mgmt port [8089]: open
New certs have been generated in 'C:\Program Files\SplunkUniversalForwarder\etc\auth'.
  Checking conf files for problems...
  Done
  Checking default conf files for edits...
  Validating installed files against hashes from 'C:\Program Files\SplunkUniversalForwarder\splunkforwarder-10.2.2-80b90d638de6-windows-x64-manifest'
  All installed files intact.
  Done
All preliminary checks passed.

Starting splunk server daemon (splunkd)...
SplunkForwarder: Unable to start the service: Access is denied.

C:\Program Files\SplunkUniversalForwarder\bin>
```

Figure 5: Starting Splunk Universal Forwarder Service

```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>splunk list forward-server
'splunk' is not recognized as an internal or external command,
operable program or batch file.

C:\Windows\system32>cd "C:\Program Files\SplunkUniversalForwarder\bin"
C:\Program Files\SplunkUniversalForwarder\bin>splunk start
The splunk daemon (splunkd) is already running.

C:\Program Files\SplunkUniversalForwarder\bin>splunk list forward-server
Your session is invalid. Please login.
Splunk username: admin
Password:
Active forwards:
  192.168.42.132:9997
Configured but inactive forwards:
  none

C:\Program Files\SplunkUniversalForwarder\bin>
```

Figure 6: Splunk Universal Forwarder Successfully Connected to Splunk Server

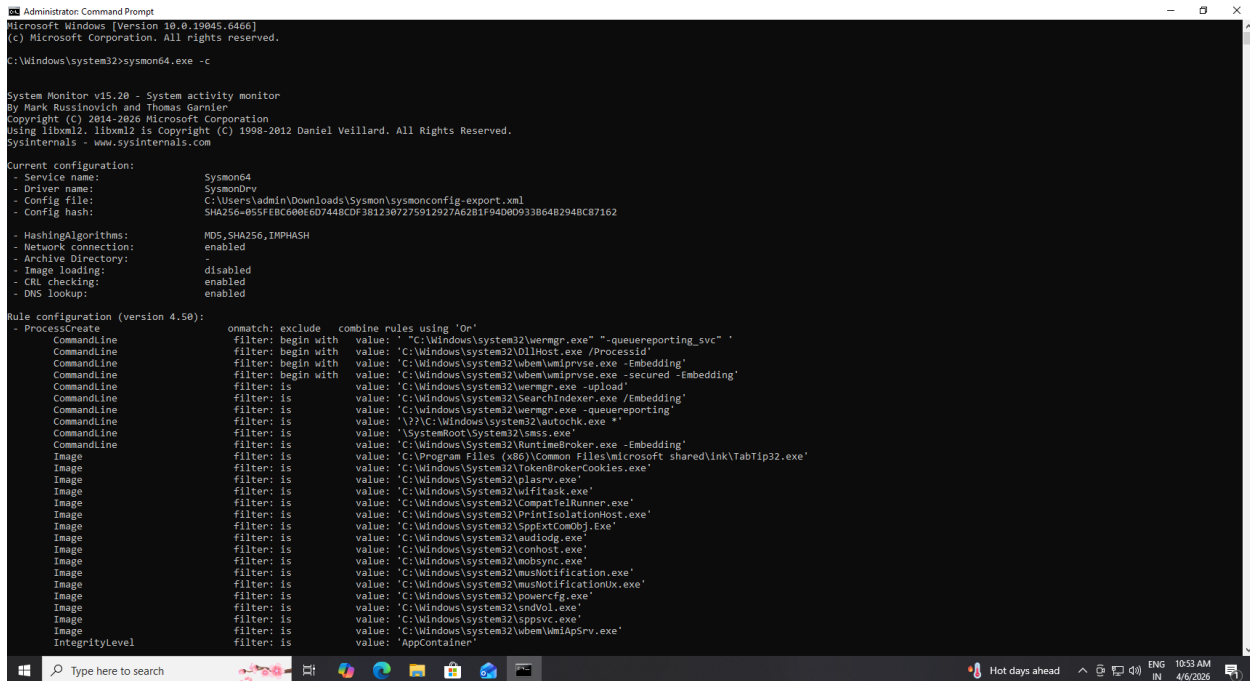
The above figure shows that the Splunk Universal Forwarder has been successfully installed on the windows machine.

The service status confirms that the forwarder is running and ready to send logs to the Splunk server.

Step 3: Configure Sysmon

Sysmon was installed with a configuration file to enable logging of:

- Process Creation (Event ID 1)
- Network connections (Event ID 3)
- Network connections (Event ID 3)

A screenshot of a Windows Command Prompt window titled 'Administrator Command Prompt'. The prompt shows the execution of 'sysmon64.exe -c' which displays the current configuration for Sysmon64. The configuration includes the service name 'Sysmon64', driver name 'SysmonDrv', and a configuration file located at 'C:\Users\admin\Downloads\Sysmon\sysmonconfig-export.xml'. It also lists the hashing algorithm as MD5, SHA256, and IMPHASH, and shows that network connections, image loading, and DNS lookups are enabled. Below the current configuration, the rule configuration (version 4.50) is displayed, showing a list of rules for ProcessCreate, CommandLine, Image, and IntegrityLevel, each with specific filters and values.

```
Administrator Command Prompt
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>sysmon64.exe -c

System Monitor v15.20 - System activity monitor
by Mark Russinovich and Thomas Garnier
Copyright (C) 2014-2020 Microsoft Corporation
Using libxml2. libxml2 is Copyright (C) 1998-2012 Daniel Veillard. All Rights Reserved.
Sysinternals - www.sysinternals.com

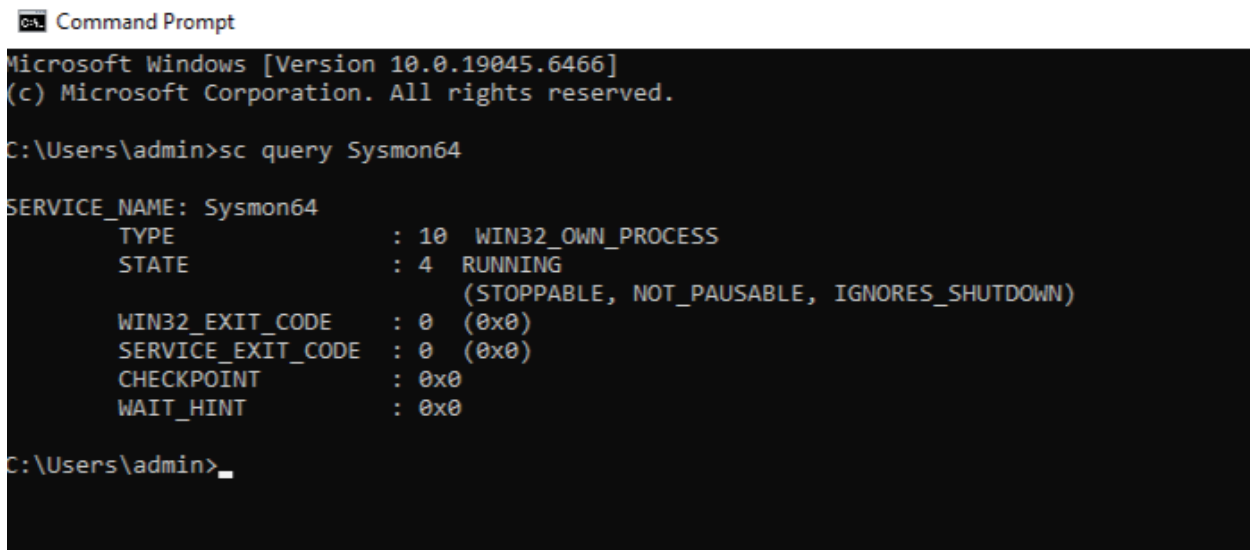
Current configuration:
- Service name: Sysmon64
- Driver name: SysmonDrv
- Config file: C:\Users\admin\Downloads\Sysmon\sysmonconfig-export.xml
- Config hash: SHA256-055F8C608E6D7448CDF3812307275912927A62B1F94D080933B6482948C87162

- HashingAlgorithms: MD5, SHA256, IMPHASH
- Network connection: enabled
- Archive Directory: disabled
- Image loading: enabled
- CRL checking: enabled
- DNS lookup: enabled

Rule configuration (version 4.50):
- ProcessCreate
  CommandLine filter: begin with value: "C:\Windows\system32\wermgr.exe" "-queue-reporting_svc"
  CommandLine filter: begin with value: "C:\Windows\system32\DllHost.exe /ProcessId"
  CommandLine filter: begin with value: "C:\Windows\system32\wbem\wmiprvse.exe -embedding"
  CommandLine filter: is value: "C:\Windows\system32\wermgr.exe -upload"
  CommandLine filter: is value: "C:\Windows\system32\SearchIndexer.exe /Embedding"
  CommandLine filter: is value: "C:\Windows\system32\wermgr.exe -queue-reporting"
  CommandLine filter: is value: "\\??\C:\Windows\system32\autochk.exe *"
  CommandLine filter: is value: "\SystemRoot\System32\smss.exe"
  CommandLine filter: is value: "C:\Windows\system32\SearchIndexer.exe /Embedding"
  Image filter: is value: "C:\Program Files (x86)\Common Files\Microsoft shared\ink\TabTip32.exe"
  Image filter: is value: "C:\Windows\System32\TokenBrokerCookies.exe"
  Image filter: is value: "C:\Windows\System32\plavrv.exe"
  Image filter: is value: "C:\Windows\System32\WlFitask.exe"
  Image filter: is value: "C:\Windows\System32\CompatTelRunner.exe"
  Image filter: is value: "C:\Windows\System32\PrintIsolationHost.exe"
  Image filter: is value: "C:\Windows\system32\SpptExtConObj.exe"
  Image filter: is value: "C:\Windows\system32\audiogd.exe"
  Image filter: is value: "C:\Windows\system32\conhost.exe"
  Image filter: is value: "C:\Windows\system32\mobsync.exe"
  Image filter: is value: "C:\Windows\system32\musnotification.exe"
  Image filter: is value: "C:\Windows\system32\musnotification.exe"
  Image filter: is value: "C:\Windows\system32\powercfg.exe"
  Image filter: is value: "C:\Windows\system32\sndvol.exe"
  Image filter: is value: "C:\Windows\system32\svchost.exe"
  Image filter: is value: "C:\Windows\system32\wbem\wmiprvse.exe"
  IntegrityLevel filter: is value: "AppContainer"

omatch: exclude combine rules using 'Or'
```

Figure 7: Sysmon Configuration Showing Enabled Logging Features

A screenshot of a Windows Command Prompt window titled 'Command Prompt'. The prompt shows the execution of 'sc query Sysmon64' which displays the service status for Sysmon64. The output shows that the service is running (STATE: 4 RUNNING) and is configured with various settings including WIN32_EXIT_CODE, SERVICE_EXIT_CODE, CHECKPOINT, and WAIT_HINT.

```
C:\Users\admin>sc query Sysmon64

SERVICE_NAME: Sysmon64
        TYPE               : 10  WIN32_OWN_PROCESS
        STATE                : 4   RUNNING
                                (STOPPABLE, NOT_PAUSABLE, IGNORES_SHUTDOWN)
        WIN32_EXIT_CODE       : 0    (0x0)
        SERVICE_EXIT_CODE    : 0    (0x0)
        CHECKPOINT            : 0x0
        WAIT_HINT             : 0x0

C:\Users\admin>
```

Figure 8: Sysmon Service Running on Windows System

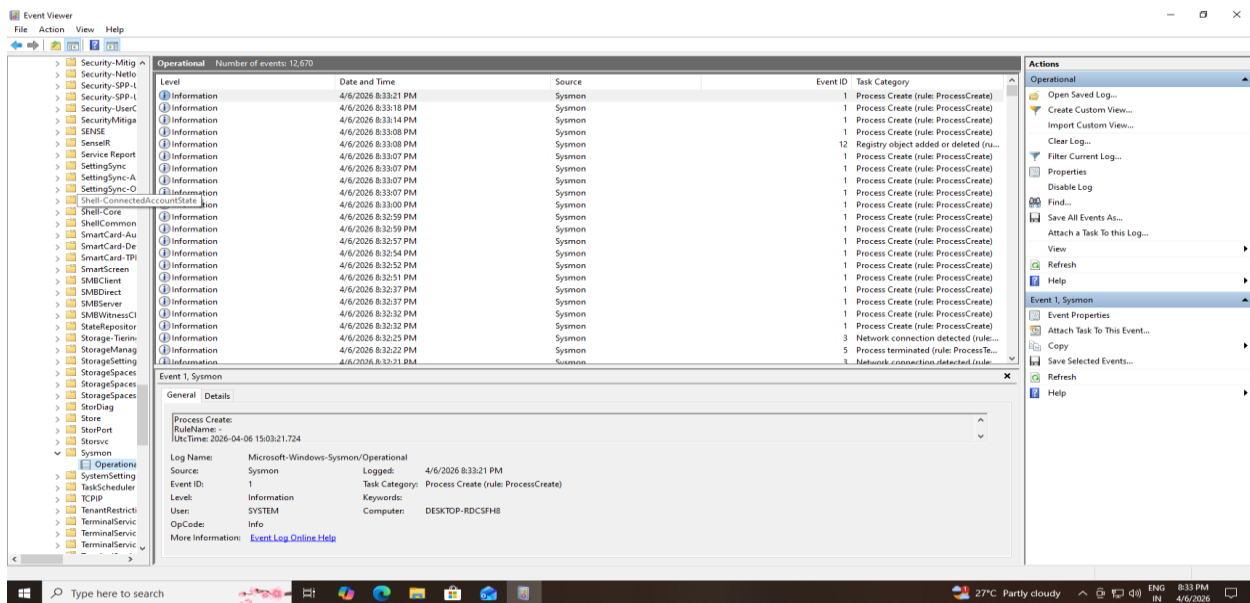


Figure 9: Sysmon logs visible in Event Viewer

The above figure shows that Sysmon has been successfully installed with the configuration file, enabling detailed system monitoring.

The configuration output confirms that network logging and other features are enabled.

The event Viewer Logs Demonstrate that Sysmon is actively recording system activity.

Step 4: Connect Forwarder to Splunk

The forwarder was configured to send logs to the Splunk server.

```
Administrator: Command Prompt
C:\Windows\system32>splunk list forward-server
'splunk' is not recognized as an internal or external command,
operable program or batch file.

C:\Windows\system32>cd "C:\Program Files\SplunkUniversalForwarder\bin" splunk start
The system cannot find the path specified.

C:\Windows\system32>cd "C:\Program Files\SplunkUniversalForwarder\bin"

C:\Program Files\SplunkUniversalForwarder\bin>splunk start
The splunk daemon (splunkd) is already running.

C:\Program Files\SplunkUniversalForwarder\bin>splunk list forward-server
Your session is invalid. Please login.
Splunk username: admin
Password:
Active forwards:
192.168.42.132:9997
Configured but inactive forwards:
None

C:\Program Files\SplunkUniversalForwarder\bin>
```

Figure 10: Forwarder Successfully Connected to Splunk Server

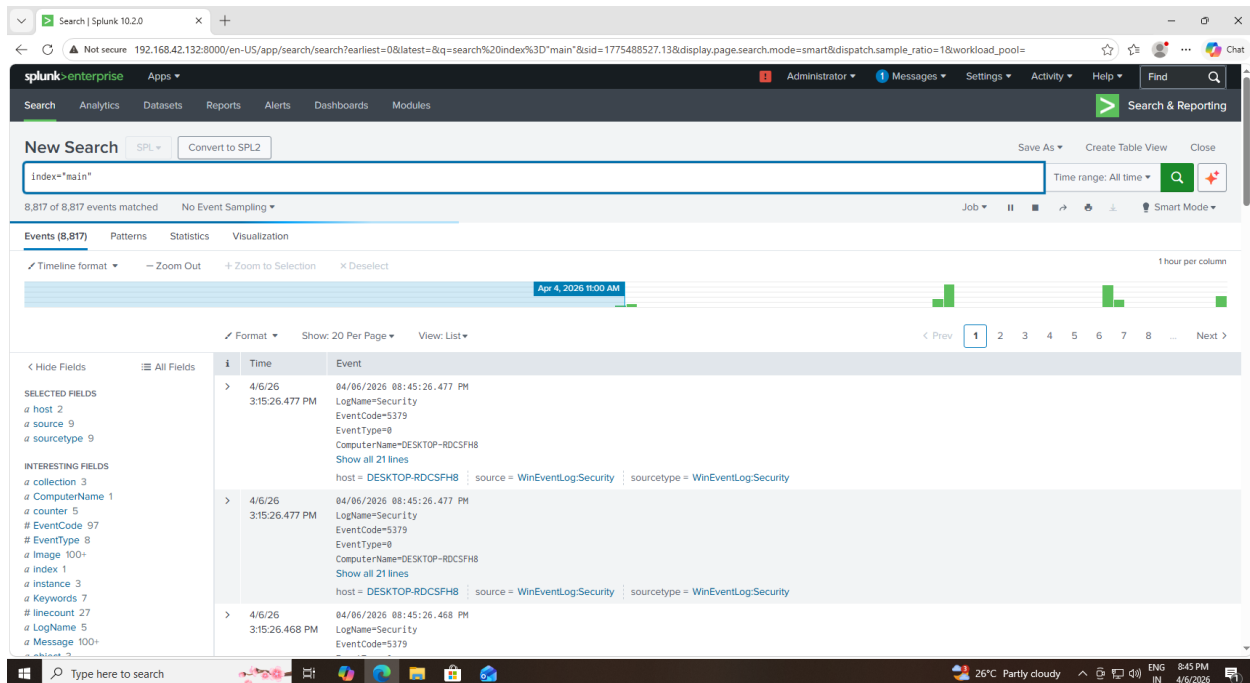


Figure 11: Logs Successfully Received in Splunk from Forwarder

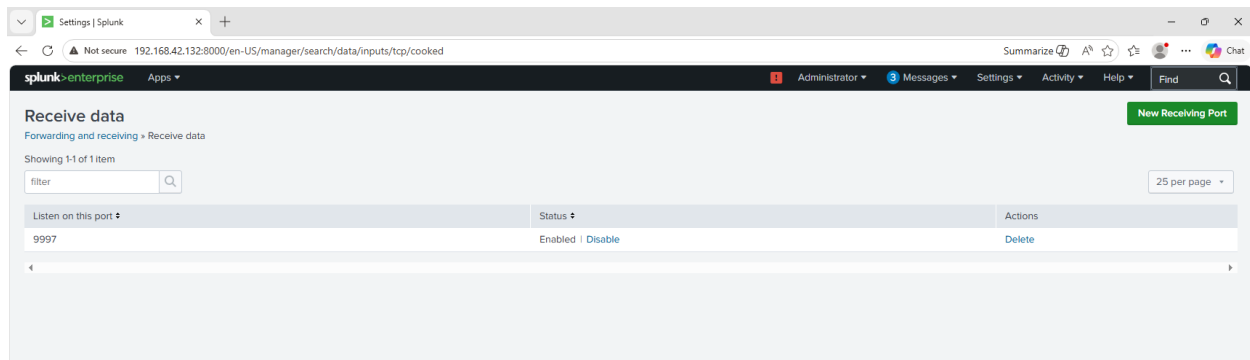


Figure 12: Splunk Server Configured to Receive Data on Port 9997

The Splunk Universal Forwarder was configured to send logs to the Splunk server using the server IP address and port 9997.

The above figure shows that the forwarder is actively connected to the Splunk server.

The logs are successfully received and displayed in Splunk, confirming proper data flow.

Step 5: Configure Log Inputs

Sysmon logs were configured in inputs.conf to be forwarded to Splunk.

```
inputs.conf - Notepad
File Edit Format View Help
[WinEventLog://Security]
disabled = 0

[WinEventLog://System]
disabled = 0

[WinEventLog://Application]
disabled = 0

[WinEventLog://Microsoft-Windows-Sysmon/Operational]
disabled = 0

[WinEventLog://Microsoft-Windows-Sysmon/Operational]
disabled = 0
index = main
```

Figure 13: Configuration of Sysmon Log Inputs in Inputs.conf File

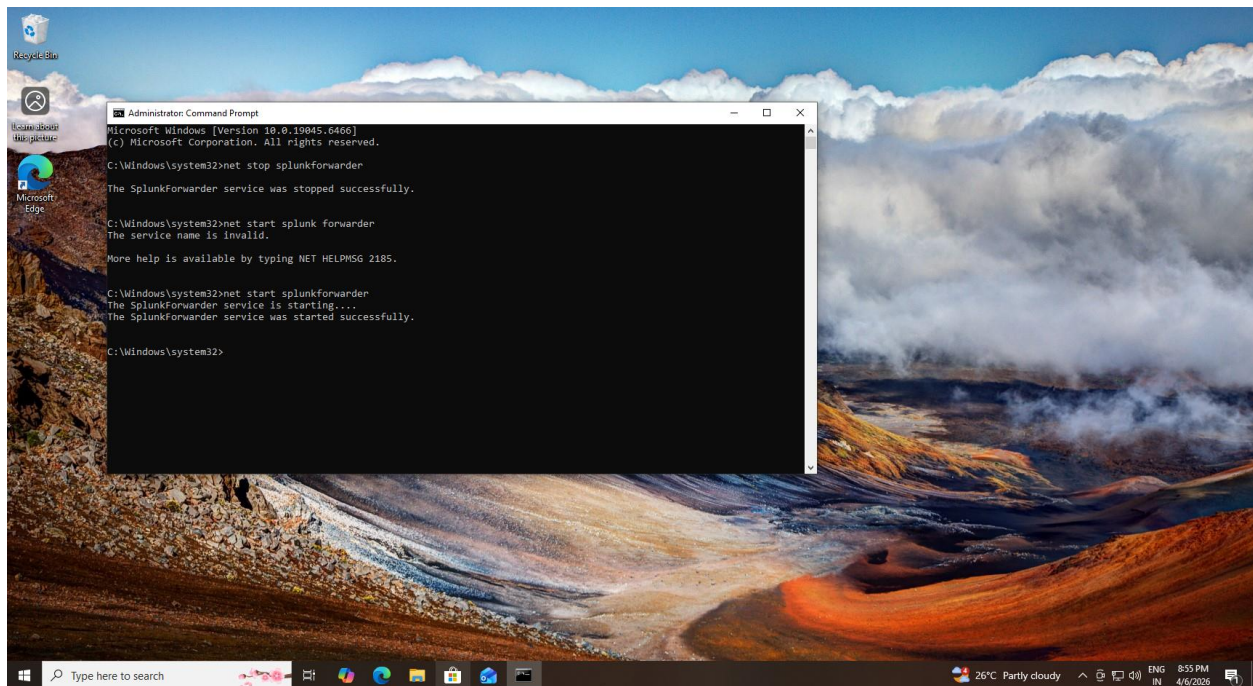


Figure 14: Restarting Splunk Universal Forwarder to Apply Log Configuration

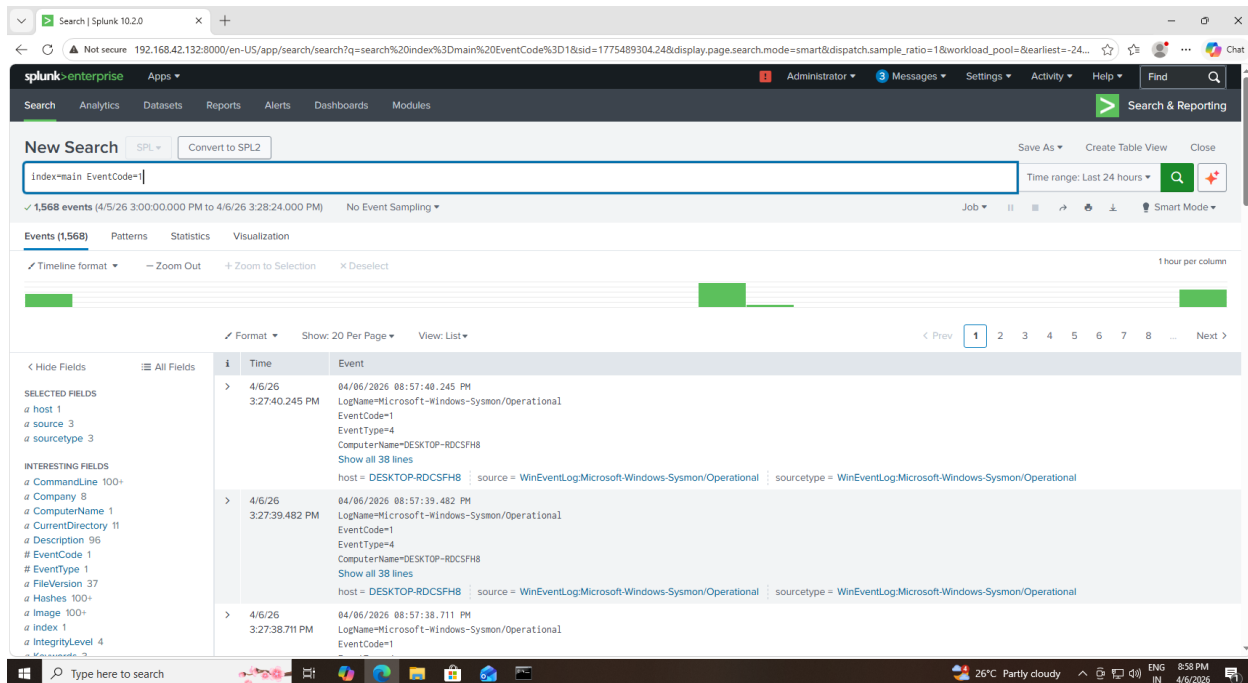


Figure 15: Sysmon Logs Successfully Collected and Visible in Splunk

The inputs.conf file was configured to collect Sysmon logs from the Windows system and forward them to the Splunk server.

The configuration ensures that Sysmon operational logs are enabled and sent to the main index.

The forwarder service was restarted to apply the new configuration settings.

The logs are successfully visible in Splunk, confirming that the input configuration is working correctly.

Step 6: Create Alerts

Alerts were created in Splunk to detect suspicious activities such as PowerShell execution.

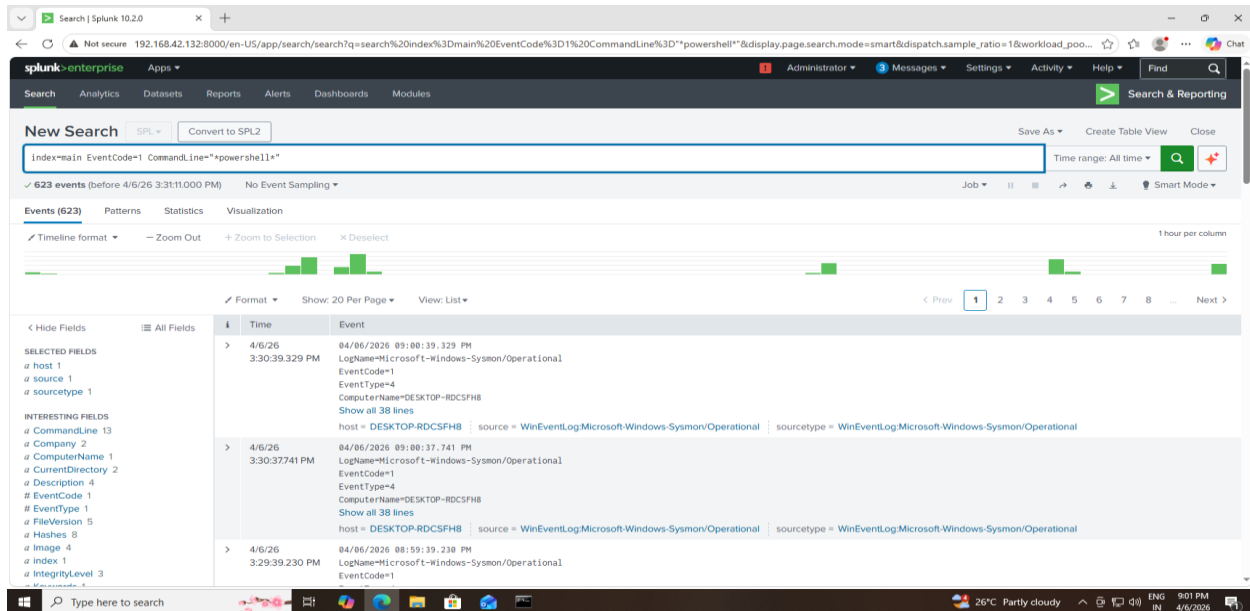


Figure 16: Detection Query for Suspicious PowerShell Execution

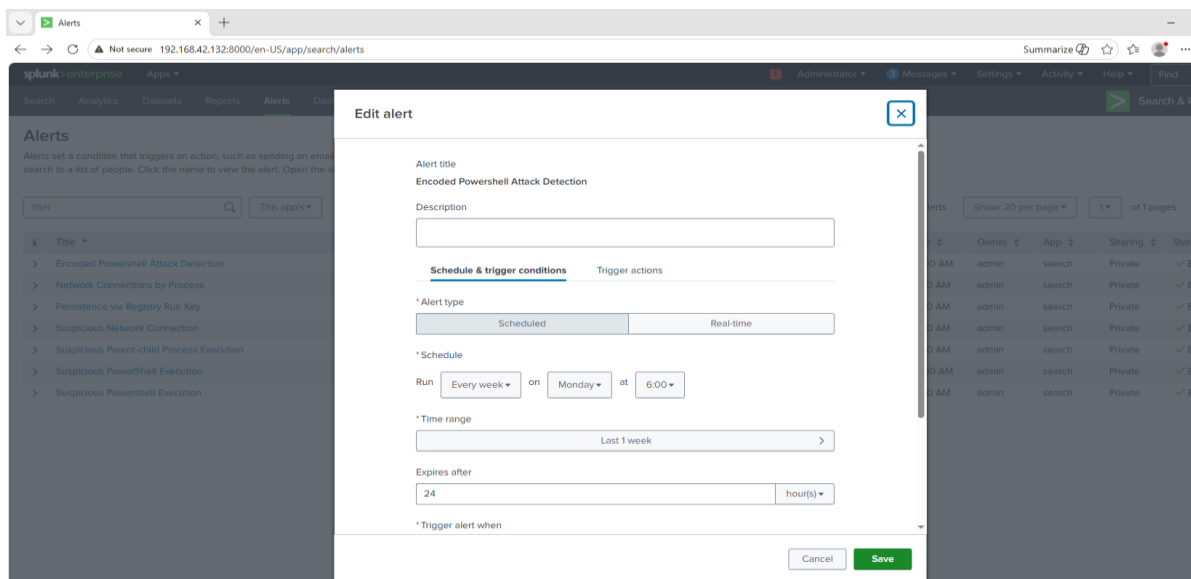


Figure 17: Configuration of Alert for Suspicious Activity Detection

Time	Alert name	App	Type	Severity	Mode	Actions
2026-04-06 15:35:30 UTC	Network Connections by Process	search	Real-time	High	Digest	View Results Edit Alert Delete
2026-04-06 15:35:25 UTC	Network Connections by Process	search	Real-time	High	Digest	View Results Edit Alert Delete
2026-04-06 15:35:20 UTC	Network Connections by Process	search	Real-time	High	Digest	View Results Edit Alert Delete
2026-04-06 15:35:20 UTC	Network Connections by Process	search	Real-time	High	Digest	View Results Edit Alert Delete
2026-04-06 15:35:15 UTC	Network Connections by Process	search	Real-time	High	Digest	View Results Edit Alert Delete
2026-04-06 15:35:10 UTC	Network Connections by Process	search	Real-time	High	Digest	View Results Edit Alert Delete
2026-04-06 15:35:10 UTC	Network Connections by Process	search	Real-time	High	Digest	View Results Edit Alert Delete
2026-04-06 15:35:05 UTC	Network Connections by Process	search	Real-time	High	Digest	View Results Edit Alert Delete
2026-04-06 15:35:00 UTC	Network Connections by Process	search	Real-time	High	Digest	View Results Edit Alert Delete
2026-04-06 15:34:55 UTC	Network Connections by Process	search	Real-time	High	Digest	View Results Edit Alert Delete
2026-04-06 15:34:50 UTC	Network Connections by Process	search	Real-time	High	Digest	View Results Edit Alert Delete
2026-04-06 15:34:50 UTC	Network Connections by Process	search	Real-time	High	Digest	View Results Edit Alert Delete
2026-04-06 15:34:45 UTC	Network Connections by Process	search	Real-time	High	Digest	View Results Edit Alert Delete
2026-04-06 15:34:40 UTC	Network Connections by Process	search	Real-time	High	Digest	View Results Edit Alert Delete
2026-04-06 15:34:40 UTC	Network Connections by Process	search	Real-time	High	Digest	View Results Edit Alert Delete
2026-04-06 15:34:40 UTC	Network Connections by Process	search	Real-time	High	Digest	View Results Edit Alert Delete
2026-04-06 15:34:39 UTC	Network Connections by Process	search	Real-time	High	Digest	View Results Edit Alert Delete

Figure 18: Alert Triggered for Suspicious PowerShell Activity

Time	Event
4/6/26 3:39:39.342 PM	04/06/2026 09:09:39.342 PM LogName=Microsoft-Windows-Sysmon/Operational EventCode=1 EventType=4 ComputerName=DESKTOP-RDCSFH8 Show all 38 lines host = DESKTOP-RDCSFH8 source = WinEventLog:Microsoft-Windows-Sysmon/Operational sourcetype = WinEventLog:Microsoft-Windows-Sysmon/Operational
4/6/26 3:39:38.589 PM	04/06/2026 09:09:38.589 PM LogName=Microsoft-Windows-Sysmon/Operational EventCode=1 EventType=4 ComputerName=DESKTOP-RDCSFH8 Show all 38 lines host = DESKTOP-RDCSFH8 source = WinEventLog:Microsoft-Windows-Sysmon/Operational sourcetype = WinEventLog:Microsoft-Windows-Sysmon/Operational
4/6/26 3:38:39.324 PM	04/06/2026 09:08:39.324 PM LogName=Microsoft-Windows-Sysmon/Operational EventCode=1 EventType=4 ComputerName=DESKTOP-RDCSFH8 Show all 38 lines host = DESKTOP-RDCSFH8 source = WinEventLog:Microsoft-Windows-Sysmon/Operational sourcetype = WinEventLog:Microsoft-Windows-Sysmon/Operational

Figure 19: Dashboard Panel Showing PowerShell Execution Activity

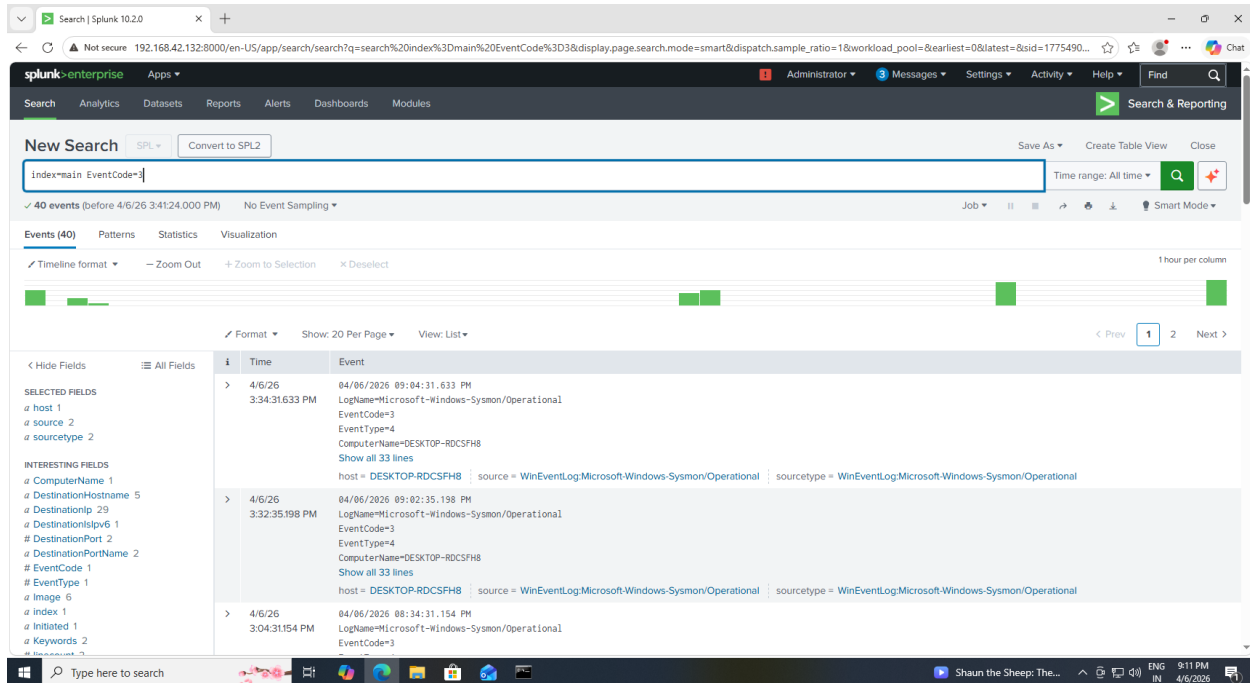


Figure 20: Dashboard Panel Showing Network Connections

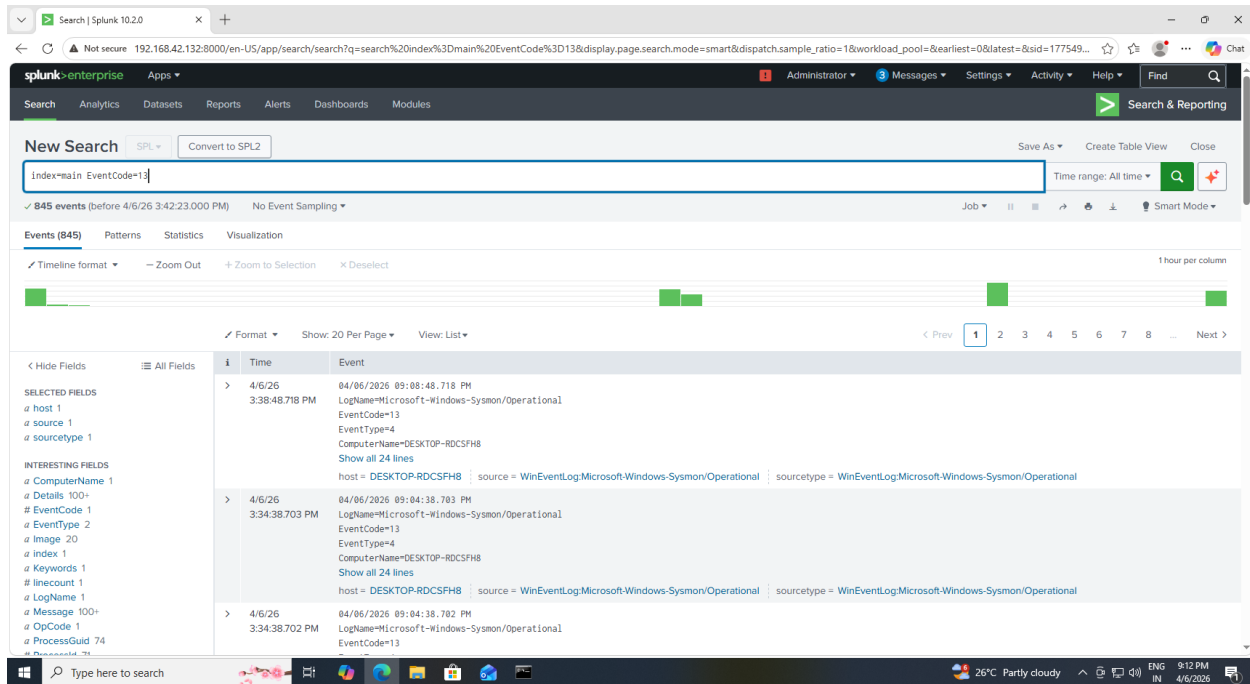


Figure 21: Dashboard Panel Showing Registry Persistence Activity

Step 7: Build Dashboard

A dashboard was created to visualize logs and monitor system activity.

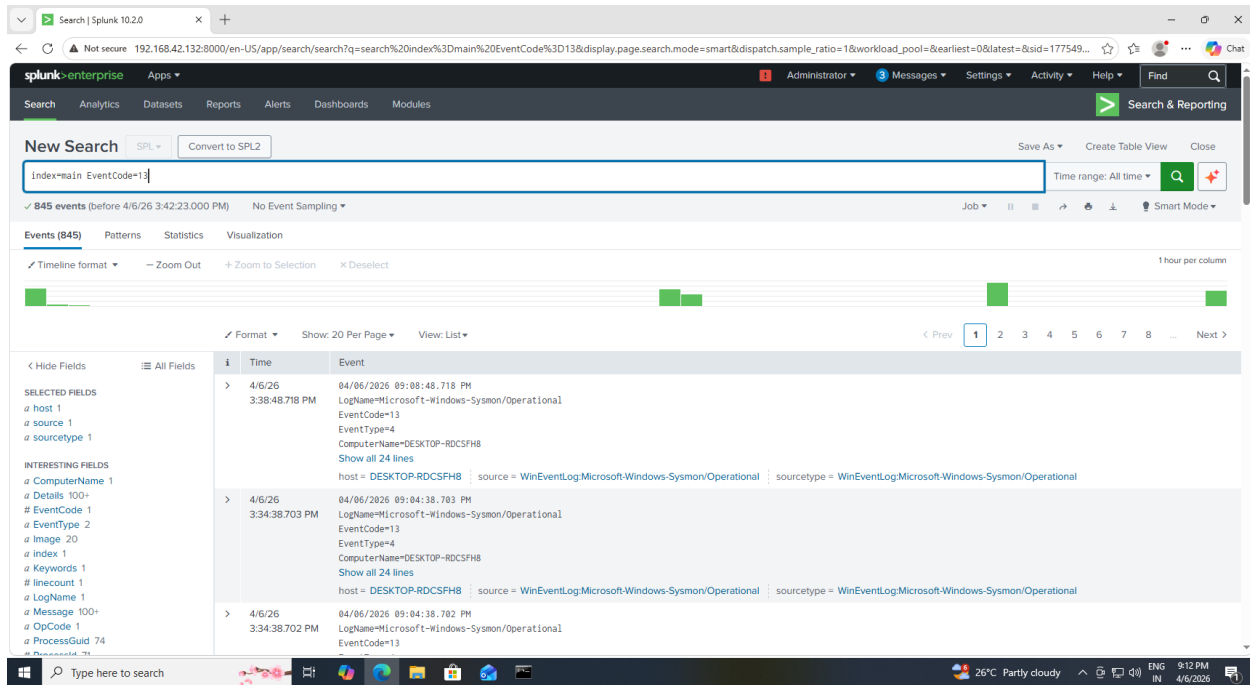


Figure 22: Splunk Dashboard Showing Real-Time Threat Monitoring

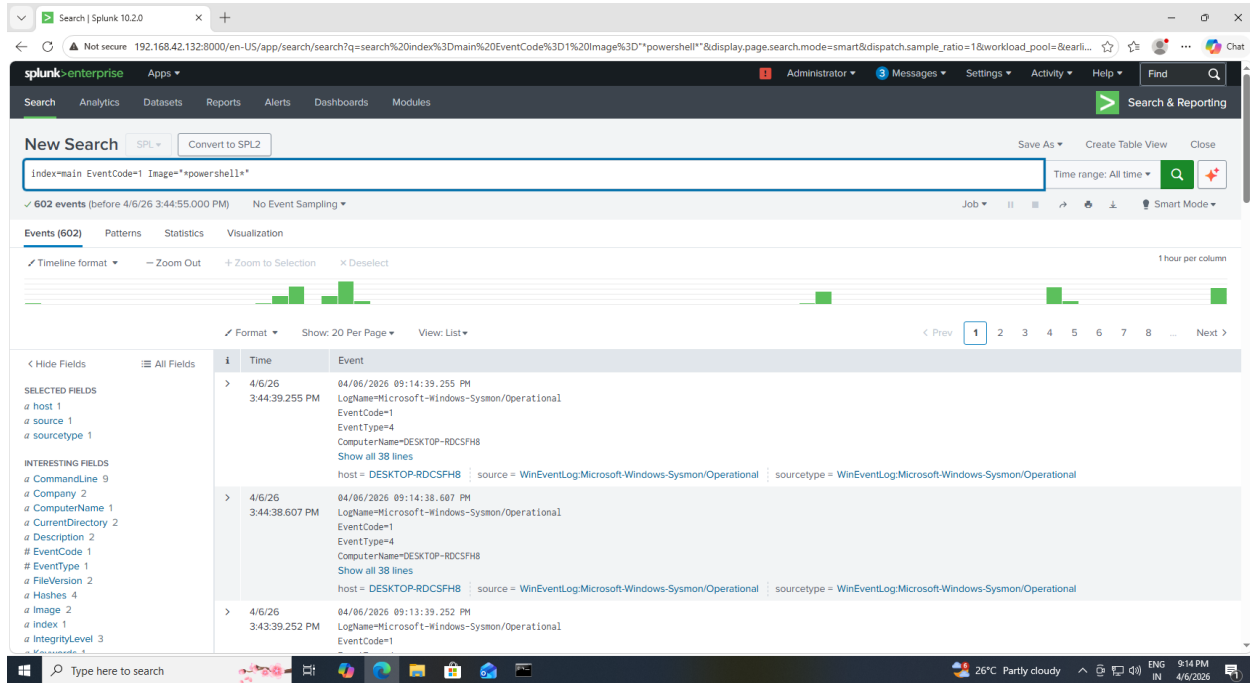


Figure 23: Dashboard Panel Displaying PowerShell Execution Activity

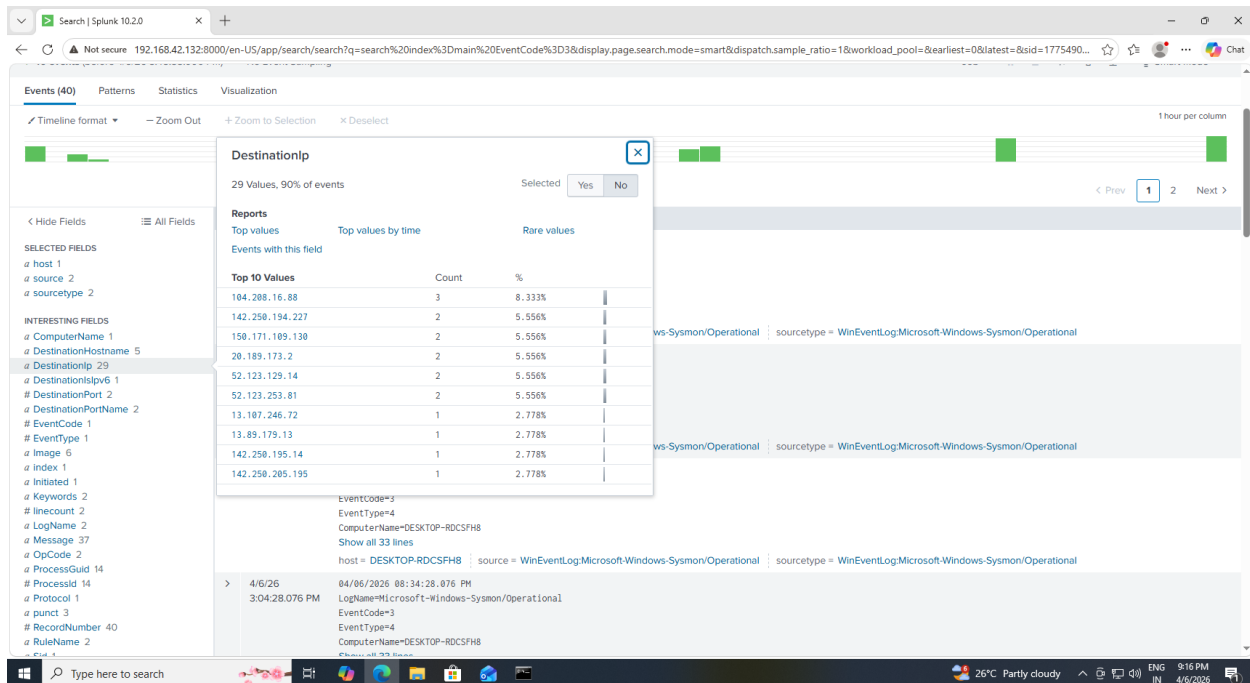


Figure 24: Dashboard Panel Showing Network Connection Activity

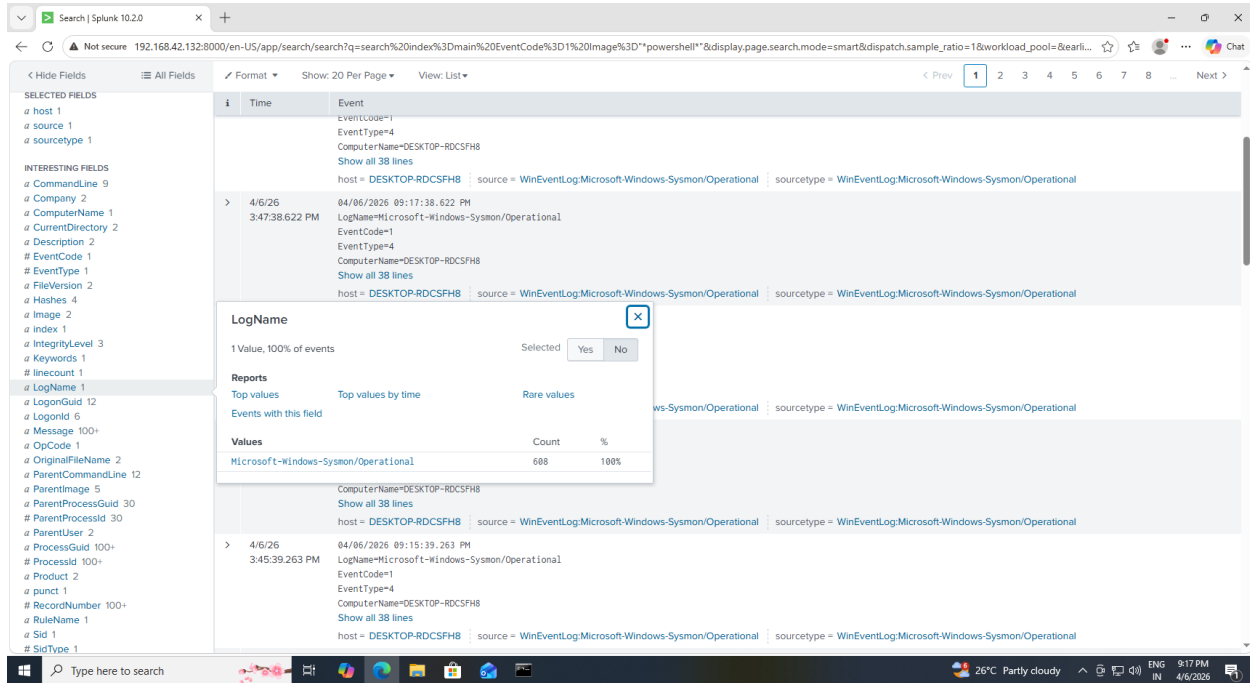


Figure 25: Dashboard Panel Displaying PowerShell Execution Activity

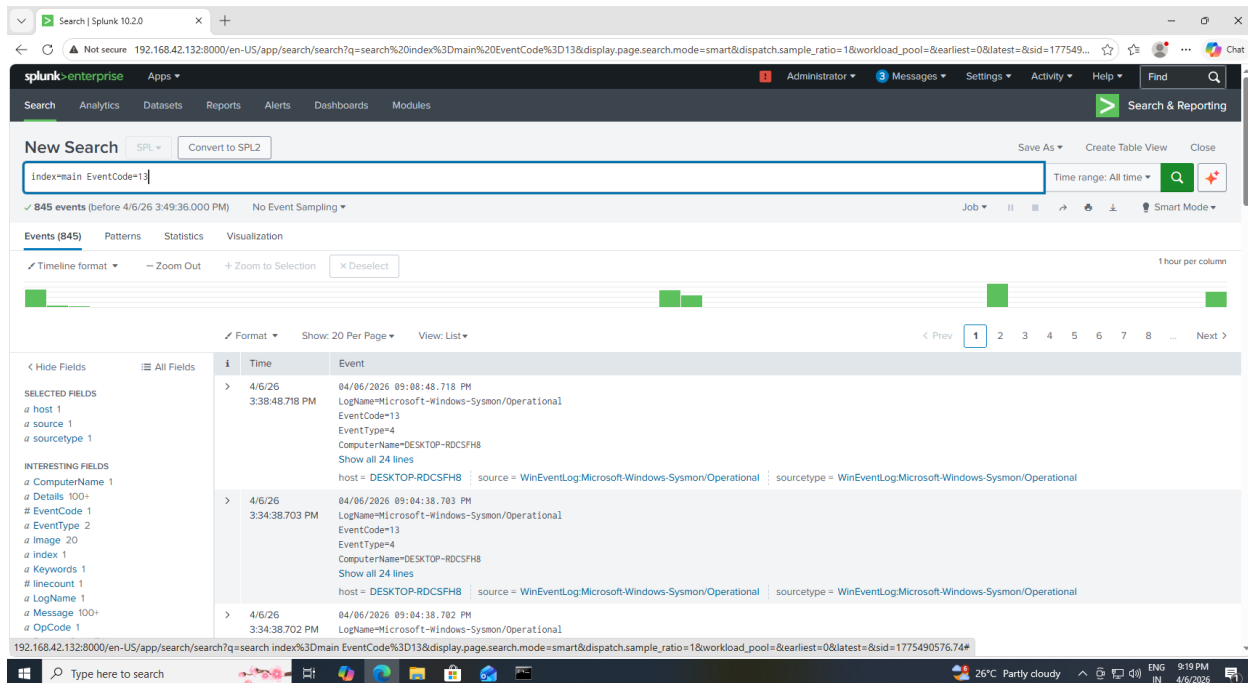


Figure 26 : Dashboard Panel Showing Registry PersistenceEvents

7. Attack Simulation

The following attack techniques were simulated:

- PowerShell Execution
- Encoded PowerShell commands
- Registry persistence using Run keys
- Network activity using ping and web requests

These actions mimic real attacker behavior.

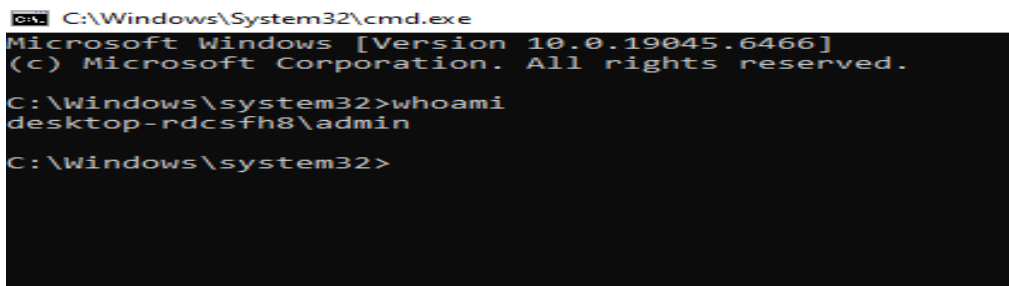


Figure 27: Normal Command Execution for Baseline Activity

```
C:\Windows\System32>cmd.exe - powershell.exe
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>whoami
desktop-rdcsh8\admin

C:\Windows\system32>powershell.exe
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Windows\system32>
```

Figure 28: Execution of PowerShell Command (Simulated Attack Behavior)

```
C:\Windows\System32>cmd.exe - powershell.exe
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>whoami
desktop-rdcsh8\admin

C:\Windows\system32>powershell.exe
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Windows\system32>powershell -EncodedCommand SQBtAGAGAYOB8AGUATABNAGUACwBMAA==
cmd.exe process the command because the value specified with -EncodedCommand is not properly encoded. The value must be Base64 encoded.

PowerShell.exe [-PSConsoleFile <file>] [-Version <version>]
  -NoLogo           Hides the copyright banner at startup.
  -NoExit           Does not exit after running startup commands.
  -STA             Starts the shell using a single-threaded apartment.
                    Single-threaded apartment (STA) is the default.
  -MTA             Start the shell using a multithreaded apartment.
  -NoProfile       Does not load the Windows PowerShell profile.
  -NonInteractive  Does not present an interactive prompt to the user.
  -InputFormat     The format of the input.

PowerShell.exe -Help | -? | ??
  Loads the specified Windows PowerShell console file. To create a console
  file, use Export-Console in Windows PowerShell.

  Starts the specified version of Windows PowerShell.
  Enter a version number with the parameter, such as "-version 2.0".

  Hides the copyright banner at startup.

  Does not exit after running startup commands.

  Starts the shell using a single-threaded apartment.
  Single-threaded apartment (STA) is the default.

  Start the shell using a multithreaded apartment.

  Does not load the Windows PowerShell profile.

  Does not present an interactive prompt to the user.

  The format of the input.
```

Figure 29: Execution of Encoded PowerShell Command (Obfuscation Techniques)

```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>reg add HKCU\Software\Microsoft\Windows\CurrentVersion\Run /v TestPersistence /t REG_SZ /d "powershell.exe" /f
The operation completed successfully.

C:\Windows\system32>
```

Figure 30: Registry Persistence Technique Simulation

```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>reg add HKCU\Software\Microsoft\Windows\CurrentVersion\Run /v TestPersistence /t REG_SZ /d "powershell.exe" /f
The operation completed successfully.

C:\Windows\system32>ping google.com

Pinging google.com [142.250.206.110] with 32 bytes of data:
Reply from 142.250.206.110: bytes=32 time=77ms TTL=128
Reply from 142.250.206.110: bytes=32 time=62ms TTL=128
Reply from 142.250.206.110: bytes=32 time=55ms TTL=128
Reply from 142.250.206.110: bytes=32 time=77ms TTL=128

Ping statistics for 142.250.206.110:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 55ms, Maximum = 77ms, Average = 67ms

C:\Windows\system32>
```

Figure 31: Network Activity Simulation

8. Detection Techniques

The following Sysmon Event IDs were used:

Event Code	Description
1	Process Creation
3	Network Connection
13	Registry Modification

Example Detection Queries:

- Detect PowerShell:

index=main EventCode=1 Image="*powershell"

- Detect Encoded Commands:

index=main EventCode=1 CommandLine="*EncodedCommand"

- Detect Persistence:

index=main EventCode=13 TargetObject="*Run"

- Detect Network Activity:

index=main EventCode=3

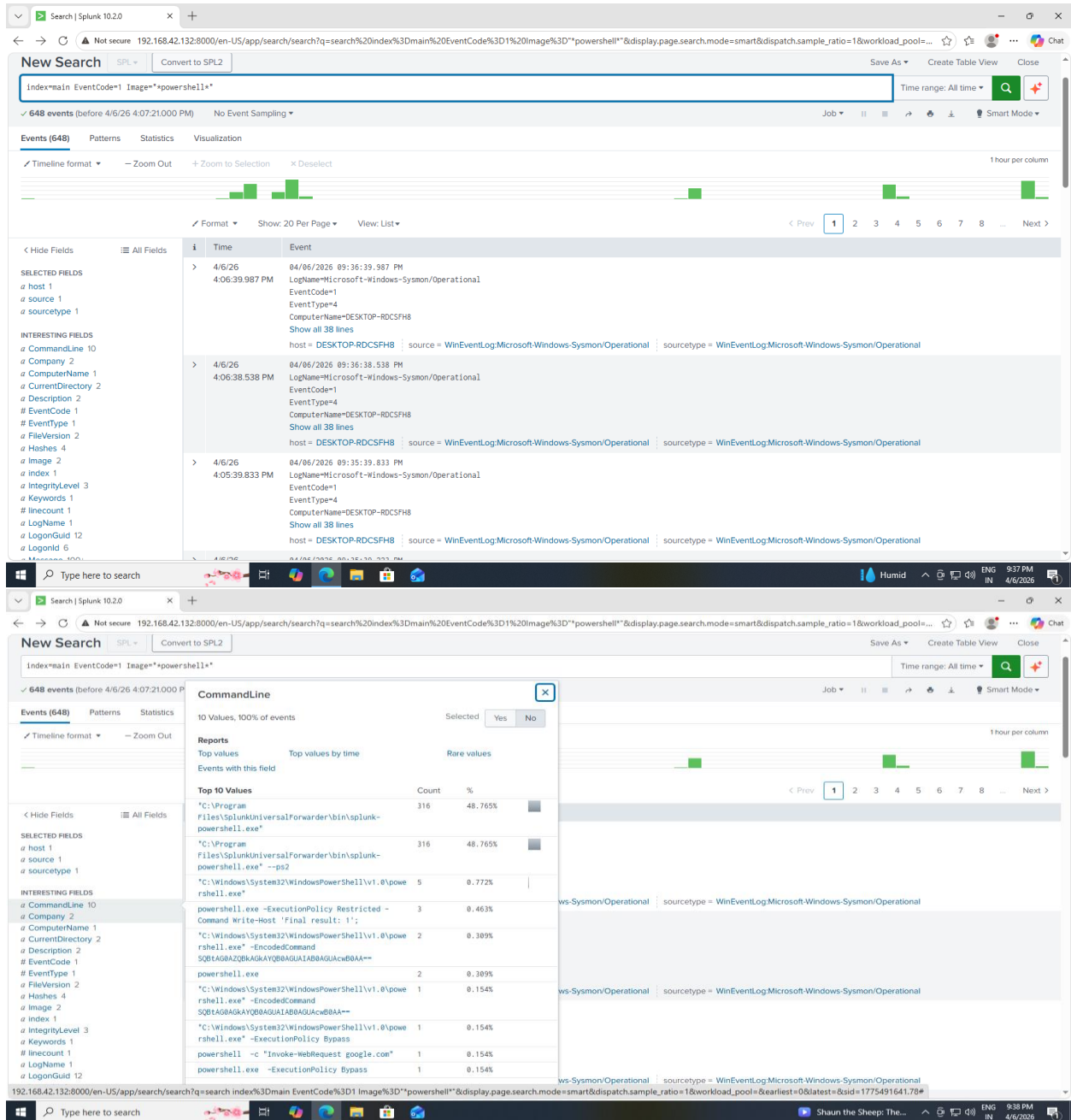


Figure 32: Detection of PowerShell Execution using EventCode 1

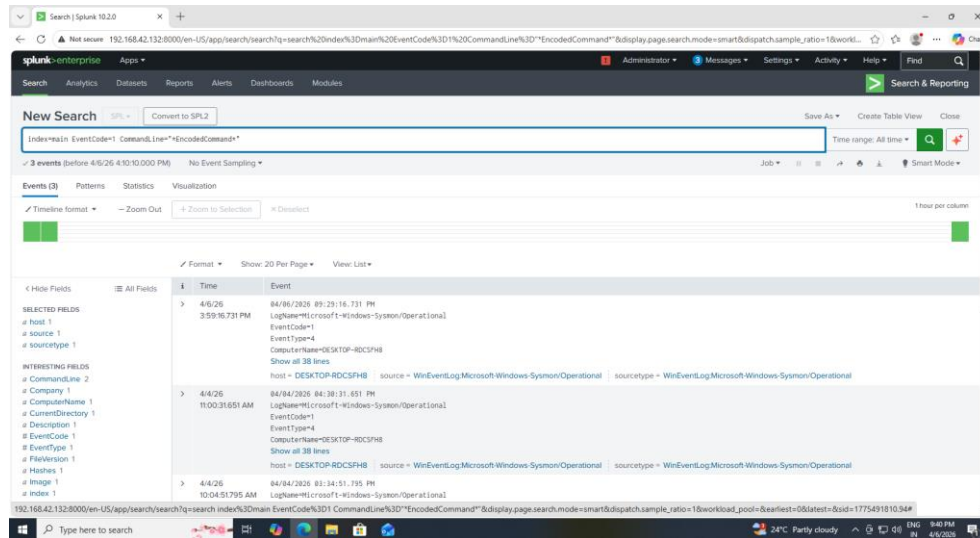


Figure 33: Detection of Encoded PowerShell Command

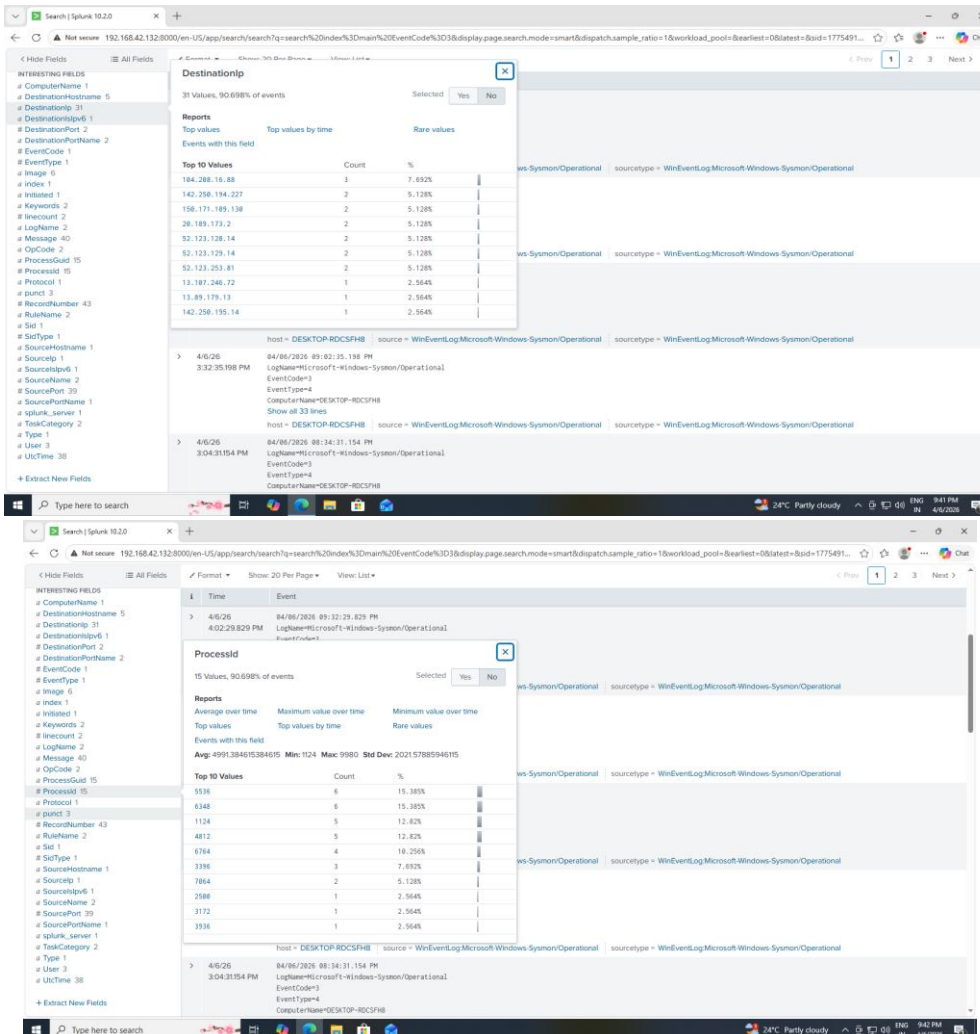


Figure 34: Detection of Network Connections using EventCode 3

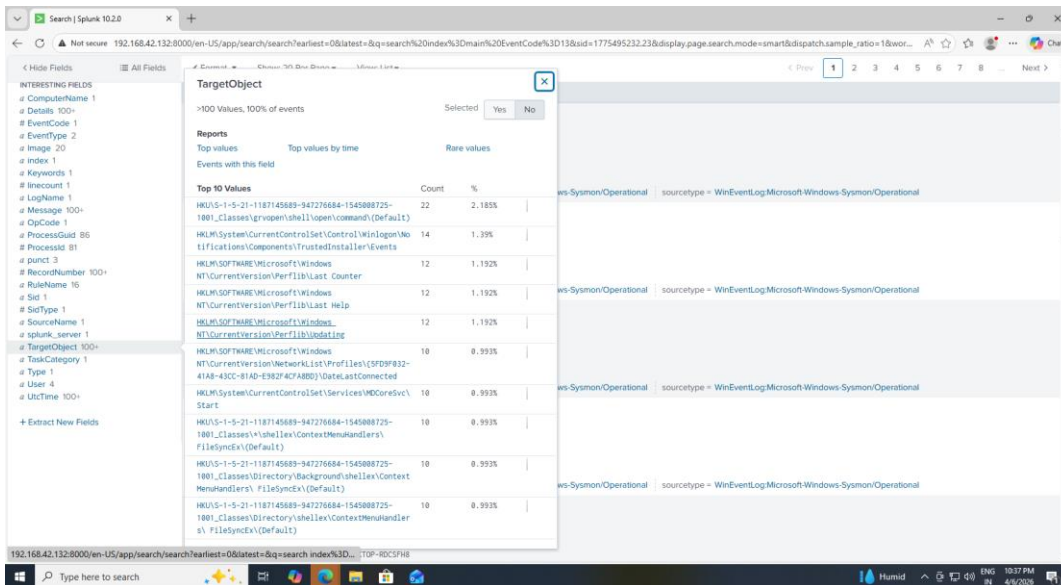


Figure 35: Detection of Registry Persistence Activity

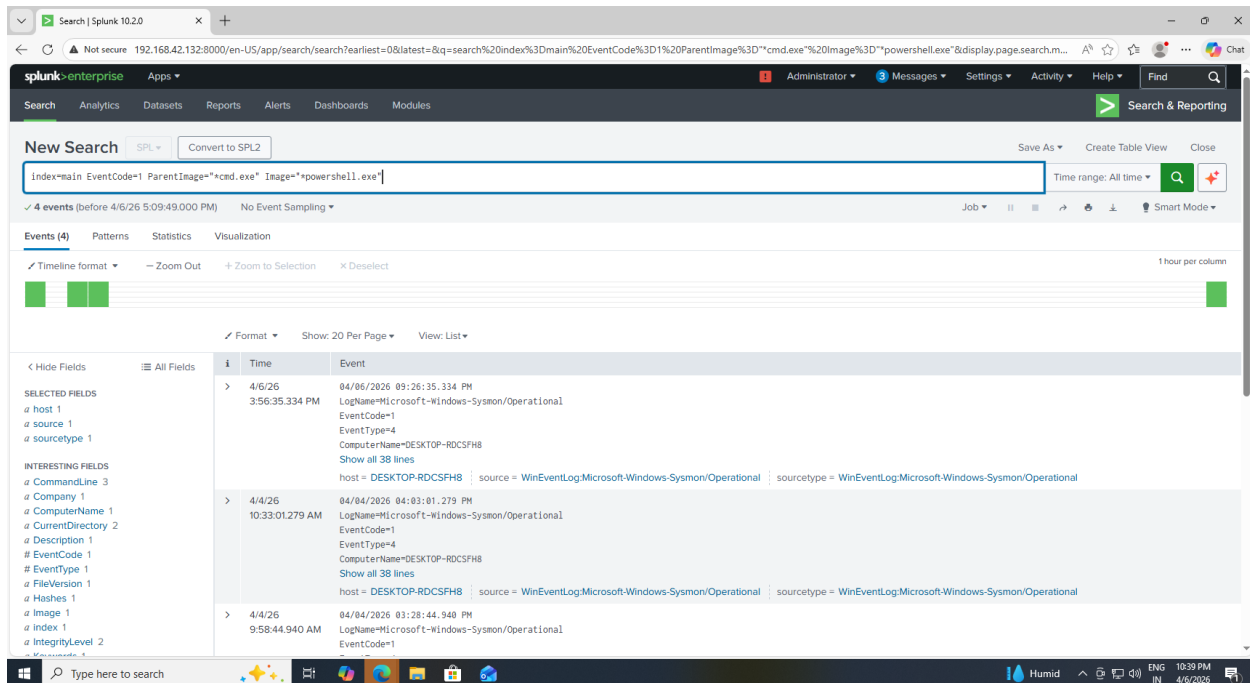


Figure 36: Detection of Suspicious Parent-Child Process Relationship

9. Dashboard

A dashboard was created in Splunk with the following panels:

The dashboard provides centralized monitoring of system activity and helps identify suspicious behavior in real time.

- PowerShell Activity
- Encoded Command Detection
- Command Prompt Activity
- Network Connections
- Registry Persistence
- Process Relationships
- Activity Timeline

The dashboard provides real-time monitoring of system behavior.

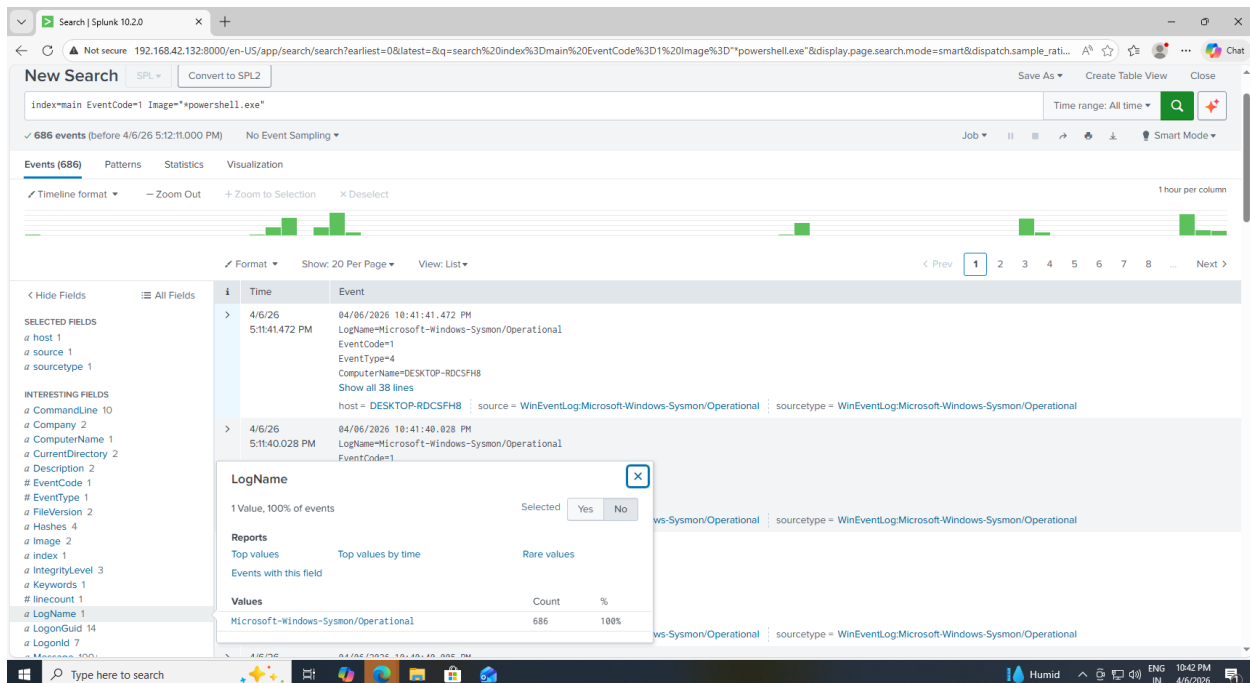


Figure 38: Dashboard Panel Showing PowerShell Execution Activity

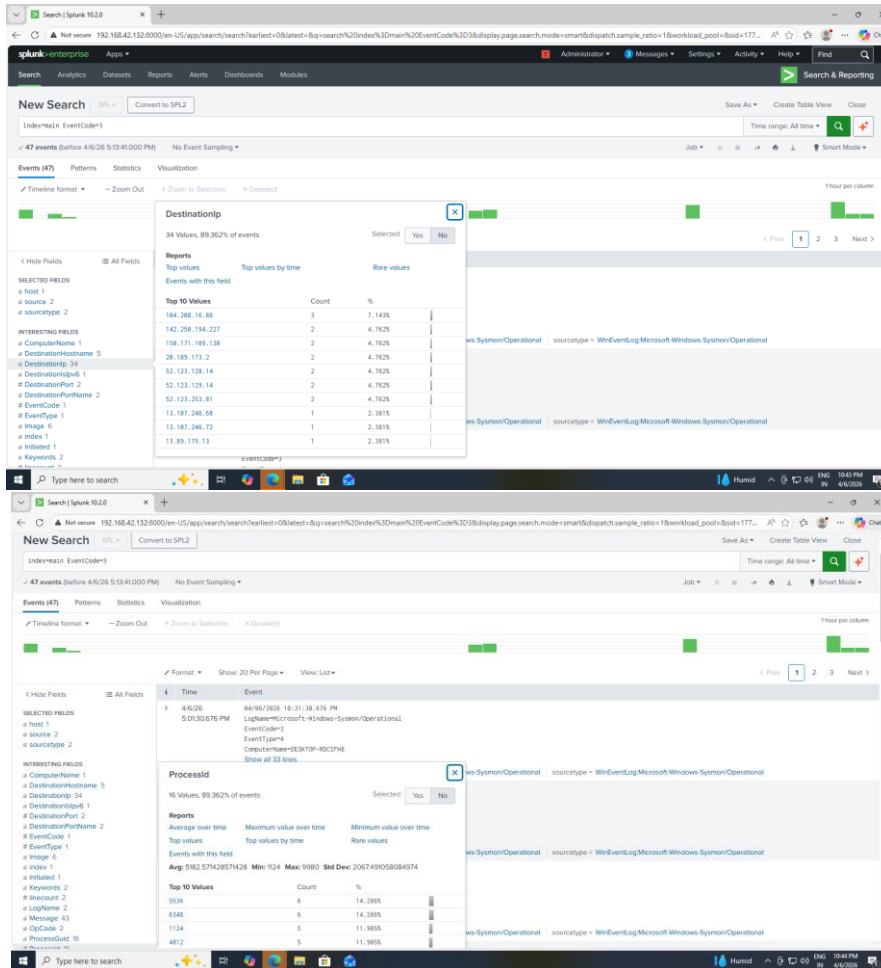


Figure 39: Dashboard Panel Showing Network Activity

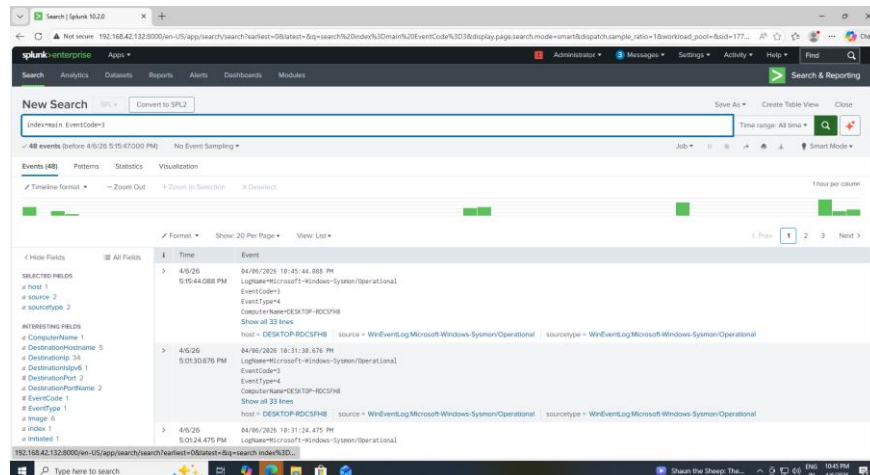


Figure 40 : Dashboard Panel Showing Registry Persistence Activity

10.Results

- The system was successfully implemented and tested using multiple attack scenarios. Logs were collected from the Windows machine and forwarded to the Splunk server in real time.
- Simulated attacks such as PowerShell execution, encoded commands, registry persistence, and network communication were successfully detected using Splunk queries.
- Alerts were triggered automatically when suspicious activity occurred, demonstrating the effectiveness of the detection system.
- The dashboard provided a clear and centralized visualization of all activities, enabling efficient monitoring and analysis.

11. Challenges Faced

- Difficulty in connecting forwarder to Splunk
- Sysmon configuration issues
- No logs appearing initially in Splunk
- Network logs not generated properly

Solutions:

- Corrected configuration files
- Restarted services
- Verified connectivity
- Applied proper Sysmon configuration

12. Conclusion

This project provided practical experience in building a SIEM system and detecting cyber attacks. It helped in understanding how attackers operate and how their behavior can be detected using logs. The use of Splunk and Sysmon proved effective in monitoring and analyzing system activity. This project enhances skills required for real-world cybersecurity roles.

13.Future Scope

- Integration with cloud-based SIEM
- Use of machine learning for threat detection
- Automated response systems
- Advanced attack simulation

14.References

- Splunk Official Documentation
- Sysmon GitHub Repository
- Microsoft Documentation
- Online Cybersecurity Tutorials